PHYDSC403T — Basic Quantum Mechanics

Unit 2: General Formalism

7th Semester, Assam University (FYUG) — Haflong Government College

Paper: PHYDSC403T, Full Marks 100 [ESE 70 + CCA 30] | Unit 2: 12 lectures

Notes 12 of 20 — drafted by gemini-3.6-flash (Kaggle Model Proxy) from the unit syllabus only, without external sources, and checked against the syllabus.

Index

Unit syllabus (as in the syllabus file)

1. Expectation value of dynamical variables, Ehrenfest theorem.
2. Stationary states, Generalized uncertainty principle.
3. Matrix representation of an operator, normalization and orthogonality of wave function in matrix form.
4. Co-ordinate and momentum representations, Wave functions in co-ordinate and momentum representations, Schrodinger wave equation in momentum representation, Momentum eigen function, Significance of momentum wave function.
5. Dirac's bra and ket notation and their operations.

Top 2 high-mark topics in this unit

1. Generalized uncertainty principle, with Heisenberg uncertainty principle — the most repeated AU topic in this unit, 4 of 6 AU questions are on uncertainty, in short and long forms.
2. Ehrenfest theorem and expectation values / stationary states — named in the course objective and unit head, derivation-heavy, and an explicit AU gap, so likely as a long question.

Contents

1. Expectation Values of Dynamical Variables
2. Ehrenfest Theorem
3. Stationary States
4. Generalised Uncertainty Principle
5. Matrix Representation of Operators
6. Coordinate and Momentum Representations
7. Dirac's Bra and Ket Notation
8. Questions — Short form, then Long form (6 AU questions)
9. Sources

1. Expectation Values of Dynamical Variables

Definition of Expectation Value

In quantum mechanics, physical observables such as position, momentum, and energy are represented by operators acting on the state function of the system. The result of a single measurement of a dynamical variable A associated with an operator $\hat{A}$ for a system in a state $\psi(x,t)$ cannot be predicted with absolute certainty; rather, quantum mechanics provides the probability distribution of possible outcomes.

The expectation value of a dynamical variable A, denoted by $\langle A \rangle$, is defined as the ensemble average or the mean value of the results of a large number of independent measurements of A performed on an ensemble of identically prepared physical systems, each described by the same normalized wave function $\psi(x,t)$.

For a normalized wave function $\psi(x,t)$ defined in one dimension, the expectation value of the operator $\hat{A}$ is given by the spatial integral:

$$\langle A \rangle = \int_{-\infty}^{\infty} \psi^*(x,t) \hat{A} \psi(x,t)\, dx$$

If the wave function $\psi(x,t)$ is not normalized to unity, the expectation value is evaluated by dividing by the norm of the wave function:

$$\langle A \rangle = \frac{\int_{-\infty}^{\infty} \psi^*(x,t) \hat{A} \psi(x,t)\, dx}{\int_{-\infty}^{\infty} \psi^*(x,t) \psi(x,t)\, dx}$$

Here, $\psi^*(x,t)$ represents the complex conjugate of the wave function $\psi(x,t)$, and the integrand $\psi^*(x,t)\hat{A}\psi(x,t)$ maintains the order of operations, as the operator $\hat{A}$ acts directly on $\psi(x,t)$.

Expectation Value of Position

The position of a particle along the x-axis is a fundamental dynamical variable. In the position representation, the position operator $\hat{x}$ acts on a wave function simply by multiplication by x:

$$\hat{x}\psi(x,t) = x\psi(x,t)$$

Substituting the position operator $\hat{x} = x$ into the general expression for the expectation value gives:

$$\langle x \rangle = \int_{-\infty}^{\infty} \psi^*(x,t) x \psi(x,t)\, dx$$

Since x is a real variable and commutes with $\psi^*(x,t)$, it can be rearranged within the integrand:

$$\langle x \rangle = \int_{-\infty}^{\infty} x \psi^*(x,t) \psi(x,t)\, dx$$

Using the definition of the probability density $P(x,t) = |\psi(x,t)|^2 = \psi^*(x,t)\psi(x,t)$, the expectation value of position simplifies to:

$$\langle x \rangle = \int_{-\infty}^{\infty} x|\psi(x,t)|^2 \, dx$$

This formula represents the standard statistical mean of the position coordinate weighted by the continuous probability density $|\psi(x,t)|^2$.

Expectation Value of Momentum

In position representation, the linear momentum operator along the x-direction is given by:

$$\hat{p} = -i\hbar\frac{\partial}{\partial x}$$

where i is the imaginary unit and $\hbar = \frac{h}{2\pi}$ is the reduced Planck constant. Substituting the momentum operator into the general definition of expectation value yields:

$$\langle p \rangle = \int_{-\infty}^{\infty} \psi^*(x,t) \left(-i\hbar\frac{\partial}{\partial x}\right) \psi(x,t) \, dx$$

Factoring out the constant multiplier $-i\hbar$ from the integral gives:

$$\langle p \rangle = -i\hbar \int_{-\infty}^{\infty} \psi^*(x,t) \frac{\partial \psi(x,t)}{\partial x} \, dx$$

This integral gives the average momentum of the quantum system across the spatial distribution described by $\psi(x,t)$.

2. Ehrenfest Theorem

Statement of Ehrenfest Theorem

Ehrenfest's theorem establishes a bridge between quantum mechanics and classical mechanics by demonstrating that the expectation values of quantum mechanical operators obey classical laws of motion.

For a particle of mass m moving in a one-dimensional potential field $V(x)$, Ehrenfest's theorem consists of two differential relations:

1. The rate of change of the expectation value of position equals the expectation value of velocity:

$$\frac{d\langle x \rangle}{dt} = \frac{\langle p \rangle}{m}$$

1. The rate of change of the expectation value of momentum equals the expectation value of the classical force acting on the particle:

$$\frac{d\langle p\rangle}{dt} = -\left\langle \frac{\partial V}{\partial x} \right\rangle$$

Proof of the First Part: $\frac{d\langle x\rangle}{dt} = \frac{\langle p\rangle}{m}$

We begin by expressing the time derivative of the expectation value of position:

$$\frac{d\langle x\rangle}{dt} = \frac{d}{dt}\int_{-\infty}^{\infty} \psi^* x \psi \, dx$$

Differentiating under the integral sign with respect to time t gives:

$$\frac{d\langle x\rangle}{dt} = \int_{-\infty}^{\infty} x \frac{\partial}{\partial t}\left(\psi^*\psi\right) \, dx$$

Applying the product rule of differentiation inside the integral yields:

$$\frac{d\langle x\rangle}{dt} = \int_{-\infty}^{\infty} x \left(\frac{\partial \psi^*}{\partial t}\psi + \psi^* \frac{\partial \psi}{\partial t} \right) \, dx$$

The time evolution of the wave function $\psi(x,t)$ is governed by the time-dependent Schrödinger equation:

$$i\hbar \frac{\partial \psi}{\partial t} = -\frac{\hbar^2}{2m}\frac{\partial^2 \psi}{\partial x^2} + V(x)\psi$$

Solving for the partial derivative of ψ with respect to time gives:

$$\frac{\partial \psi}{\partial t} = \frac{i\hbar}{2m}\frac{\partial^2 \psi}{\partial x^2} - \frac{i}{\hbar}V(x)\psi$$

Taking the complex conjugate of the time-dependent Schrödinger equation gives:

$$-i\hbar \frac{\partial \psi^*}{\partial t} = -\frac{\hbar^2}{2m}\frac{\partial^2 \psi^*}{\partial x^2} + V(x)\psi^*$$

Solving for the partial derivative of ψ^* with respect to time gives:

$$\frac{\partial \psi^*}{\partial t} = -\frac{i\hbar}{2m}\frac{\partial^2 \psi^*}{\partial x^2} + \frac{i}{\hbar}V(x)\psi^*$$

Substituting the expressions for $\frac{\partial \psi}{\partial t}$ and $\frac{\partial \psi^*}{\partial t}$ into the integral yields:

$$\frac{d\langle x\rangle}{dt} = \int_{-\infty}^{\infty} x \left[\left(-\frac{i\hbar}{2m}\frac{\partial^2 \psi^*}{\partial x^2} + \frac{i}{\hbar}V\psi^* \right)\psi + \psi^* \left(\frac{i\hbar}{2m}\frac{\partial^2 \psi}{\partial x^2} - \frac{i}{\hbar}V\psi \right) \right] \, dx$$

Expanding the terms inside the brackets yields:

$$\frac{d\langle x\rangle}{dt} = \int_{-\infty}^{\infty} x \left(-\frac{i\hbar}{2m}\frac{\partial^2 \psi^*}{\partial x^2}\psi + \frac{i}{\hbar}V\psi^*\psi + \frac{i\hbar}{2m}\psi^*\frac{\partial^2 \psi}{\partial x^2} - \frac{i}{\hbar}V\psi^*\psi \right) \, dx$$

The terms involving the potential $V(x)$ cancel out, leaving:

$$\frac{d\langle x\rangle}{dt} = \frac{i\hbar}{2m}\int_{-\infty}^{\infty} x\left(\psi^*\frac{\partial^2\psi}{\partial x^2} - \frac{\partial^2\psi^*}{\partial x^2}\psi\right)\,dx$$

The integrand term inside the parentheses can be written as a spatial derivative:

$$\psi^*\frac{\partial^2\psi}{\partial x^2} - \frac{\partial^2\psi^*}{\partial x^2}\psi = \frac{\partial}{\partial x}\left(\psi^*\frac{\partial\psi}{\partial x} - \frac{\partial\psi^*}{\partial x}\psi\right)$$

Substituting this identity into the integral gives:

$$\frac{d\langle x\rangle}{dt} = \frac{i\hbar}{2m}\int_{-\infty}^{\infty} x\frac{\partial}{\partial x}\left(\psi^*\frac{\partial\psi}{\partial x} - \frac{\partial\psi^*}{\partial x}\psi\right)\,dx$$

Integrating by parts with $u = x$ and $dv = \frac{\partial}{\partial x}\left(\psi^*\frac{\partial\psi}{\partial x} - \frac{\partial\psi^*}{\partial x}\psi\right)dx$ yields:

$$\frac{d\langle x\rangle}{dt} = \frac{i\hbar}{2m}\left[x\left(\psi^*\frac{\partial\psi}{\partial x} - \frac{\partial\psi^*}{\partial x}\psi\right)\right]_{-\infty}^{\infty} - \frac{i\hbar}{2m}\int_{-\infty}^{\infty}\left(\psi^*\frac{\partial\psi}{\partial x} - \frac{\partial\psi^*}{\partial x}\psi\right)\,dx$$

For physically admissible square-integrable wave functions, $\psi(x,t)$ and its spatial derivatives vanish as $x \to \pm\infty$, causing the boundary term to evaluate to zero:

$$\frac{d\langle x\rangle}{dt} = -\frac{i\hbar}{2m}\int_{-\infty}^{\infty}\left(\psi^*\frac{\partial\psi}{\partial x} - \frac{\partial\psi^*}{\partial x}\psi\right)\,dx$$

Splitting the integral into two separate terms yields:

$$\frac{d\langle x\rangle}{dt} = -\frac{i\hbar}{2m}\int_{-\infty}^{\infty}\psi^*\frac{\partial\psi}{\partial x}\,dx + \frac{i\hbar}{2m}\int_{-\infty}^{\infty}\frac{\partial\psi^*}{\partial x}\psi\,dx$$

Integrating the second integral by parts gives:

$$\int_{-\infty}^{\infty}\frac{\partial\psi^*}{\partial x}\psi\,dx = [\psi^*\psi]_{-\infty}^{\infty} - \int_{-\infty}^{\infty}\psi^*\frac{\partial\psi}{\partial x}\,dx$$

Since the boundary term $[\psi^*\psi]_{-\infty}^{\infty}$ vanishes at infinity, we obtain:

$$\int_{-\infty}^{\infty}\frac{\partial\psi^*}{\partial x}\psi\,dx = -\int_{-\infty}^{\infty}\psi^*\frac{\partial\psi}{\partial x}\,dx$$

Substituting this result back into the equation for $\frac{d\langle x\rangle}{dt}$ gives:

$$\frac{d\langle x\rangle}{dt} = -\frac{i\hbar}{2m}\int_{-\infty}^{\infty}\psi^*\frac{\partial\psi}{\partial x}\,dx - \frac{i\hbar}{2m}\int_{-\infty}^{\infty}\psi^*\frac{\partial\psi}{\partial x}\,dx$$

Combining the identical integrals leads to:

$$\frac{d\langle x\rangle}{dt} = -\frac{i\hbar}{m}\int_{-\infty}^{\infty}\psi^*\frac{\partial\psi}{\partial x}\,dx$$

Moving the factor $-i\hbar$ inside the integral gives:

$$\frac{d\langle x\rangle}{dt} = \frac{1}{m}\int_{-\infty}^{\infty}\psi^*\left(-i\hbar\frac{\partial}{\partial x}\right)\psi\,dx$$

Recognizing the definition of the momentum expectation value $\langle p\rangle = \int_{-\infty}^{\infty}\psi^*\left(-i\hbar\frac{\partial}{\partial x}\right)\psi\,dx$, we arrive at the first relation:

$$\frac{d\langle x\rangle}{dt} = \frac{\langle p\rangle}{m}$$

Proof of the Second Part: $\frac{d\langle p\rangle}{dt} = -\left\langle\frac{\partial V}{\partial x}\right\rangle$

We start with the time derivative of the expectation value of momentum:

$$\frac{d\langle p\rangle}{dt} = \frac{d}{dt}\int_{-\infty}^{\infty}\psi^*\left(-i\hbar\frac{\partial\psi}{\partial x}\right)\,dx$$

Differentiating under the integral sign gives:

$$\frac{d\langle p\rangle}{dt} = -i\hbar\int_{-\infty}^{\infty}\left(\frac{\partial\psi^*}{\partial t}\frac{\partial\psi}{\partial x} + \psi^*\frac{\partial^2\psi}{\partial t\partial x}\right)\,dx$$

Interchanging the order of partial differentiation such that $\frac{\partial^2\psi}{\partial t\partial x} = \frac{\partial}{\partial x}\left(\frac{\partial\psi}{\partial t}\right)$ gives:

$$\frac{d\langle p\rangle}{dt} = -i\hbar\int_{-\infty}^{\infty}\frac{\partial\psi^*}{\partial t}\frac{\partial\psi}{\partial x}\,dx - i\hbar\int_{-\infty}^{\infty}\psi^*\frac{\partial}{\partial x}\left(\frac{\partial\psi}{\partial t}\right)\,dx$$

Integrating the second integral by parts yields:

$$\int_{-\infty}^{\infty}\psi^*\frac{\partial}{\partial x}\left(\frac{\partial\psi}{\partial t}\right)\,dx = \left[\psi^*\frac{\partial\psi}{\partial t}\right]_{-\infty}^{\infty} - \int_{-\infty}^{\infty}\frac{\partial\psi^*}{\partial x}\frac{\partial\psi}{\partial t}\,dx$$

Since the wave function vanishes at spatial infinity, the boundary term vanishes, giving:

$$\int_{-\infty}^{\infty}\psi^*\frac{\partial}{\partial x}\left(\frac{\partial\psi}{\partial t}\right)\,dx = -\int_{-\infty}^{\infty}\frac{\partial\psi^*}{\partial x}\frac{\partial\psi}{\partial t}\,dx$$

Substituting this result back into the expression for $\frac{d\langle p\rangle}{dt}$ yields:

$$\frac{d\langle p\rangle}{dt} = -i\hbar\int_{-\infty}^{\infty}\frac{\partial\psi^*}{\partial t}\frac{\partial\psi}{\partial x}\,dx + i\hbar\int_{-\infty}^{\infty}\frac{\partial\psi^*}{\partial x}\frac{\partial\psi}{\partial t}\,dx$$

Recall from the time-dependent Schrödinger equation and its complex conjugate:

$$\frac{\partial \psi}{\partial t} = \frac{i\hbar}{2m}\frac{\partial^2 \psi}{\partial x^2} - \frac{i}{\hbar}V\psi$$

$$\frac{\partial \psi^*}{\partial t} = -\frac{i\hbar}{2m}\frac{\partial^2 \psi^*}{\partial x^2} + \frac{i}{\hbar}V\psi^*$$

Substituting $\frac{\partial \psi^*}{\partial t}$ into the first term of the integrand gives:

$$-i\hbar\frac{\partial \psi^*}{\partial t}\frac{\partial \psi}{\partial x} = -i\hbar\left(-\frac{i\hbar}{2m}\frac{\partial^2 \psi^*}{\partial x^2} + \frac{i}{\hbar}V\psi^*\right)\frac{\partial \psi}{\partial x} = -\frac{\hbar^2}{2m}\frac{\partial^2 \psi^*}{\partial x^2}\frac{\partial \psi}{\partial x} + V\psi^*\frac{\partial \psi}{\partial x}$$

Substituting $\frac{\partial \psi}{\partial t}$ into the second term of the integrand gives:

$$i\hbar\frac{\partial \psi^*}{\partial x}\frac{\partial \psi}{\partial t} = i\hbar\frac{\partial \psi^*}{\partial x}\left(\frac{i\hbar}{2m}\frac{\partial^2 \psi}{\partial x^2} - \frac{i}{\hbar}V\psi\right) = -\frac{\hbar^2}{2m}\frac{\partial \psi^*}{\partial x}\frac{\partial^2 \psi}{\partial x^2} + V\psi\frac{\partial \psi^*}{\partial x}$$

Combining the two integrands into a single expression yields:

$$\frac{d\langle p\rangle}{dt} = \int_{-\infty}^{\infty}\left(-\frac{\hbar^2}{2m}\frac{\partial^2 \psi^*}{\partial x^2}\frac{\partial \psi}{\partial x} - \frac{\hbar^2}{2m}\frac{\partial \psi^*}{\partial x}\frac{\partial^2 \psi}{\partial x^2} + V\psi^*\frac{\partial \psi}{\partial x} + V\psi\frac{\partial \psi^*}{\partial x}\right)dx$$

Grouping the terms proportional to $-\frac{\hbar^2}{2m}$ gives:

$$-\frac{\hbar^2}{2m}\int_{-\infty}^{\infty}\left(\frac{\partial^2 \psi^*}{\partial x^2}\frac{\partial \psi}{\partial x} + \frac{\partial \psi^*}{\partial x}\frac{\partial^2 \psi}{\partial x^2}\right)dx = -\frac{\hbar^2}{2m}\int_{-\infty}^{\infty}\frac{\partial}{\partial x}\left(\frac{\partial \psi^*}{\partial x}\frac{\partial \psi}{\partial x}\right)dx$$

Integrating this exact spatial derivative yields:

$$-\frac{\hbar^2}{2m}\left[\frac{\partial \psi^*}{\partial x}\frac{\partial \psi}{\partial x}\right]_{-\infty}^{\infty} = 0$$

because the spatial derivatives of ψ vanish at infinity. Consequently, only the potential terms survive:

$$\frac{d\langle p\rangle}{dt} = \int_{-\infty}^{\infty}\left(V\psi^*\frac{\partial \psi}{\partial x} + V\psi\frac{\partial \psi^*}{\partial x}\right)dx$$

We re-evaluate the second term $V\psi\frac{\partial \psi^*}{\partial x}$ using integration by parts:

$$\int_{-\infty}^{\infty}V\psi\frac{\partial \psi^*}{\partial x}\,dx = \left[V\psi\psi^*\right]_{-\infty}^{\infty} - \int_{-\infty}^{\infty}\psi^*\frac{\partial}{\partial x}(V\psi)\,dx$$

The boundary term $\left[V\psi\psi^*\right]_{-\infty}^{\infty}$ vanishes at infinity, leaving:

$$\int_{-\infty}^{\infty}V\psi\frac{\partial \psi^*}{\partial x}\,dx = -\int_{-\infty}^{\infty}\psi^*\left(\frac{\partial V}{\partial x}\psi + V\frac{\partial \psi}{\partial x}\right)dx$$

Expanding this integral gives:

$$\int_{-\infty}^{\infty} V\psi \frac{\partial \psi^*}{\partial x}\, dx = -\int_{-\infty}^{\infty} \psi^* \frac{\partial V}{\partial x}\psi\, dx - \int_{-\infty}^{\infty} V\psi^* \frac{\partial \psi}{\partial x}\, dx$$

Substituting this result back into the formula for $\frac{d\langle p\rangle}{dt}$ gives:

$$\frac{d\langle p\rangle}{dt} = \int_{-\infty}^{\infty} V\psi^* \frac{\partial \psi}{\partial x}\, dx - \int_{-\infty}^{\infty} \psi^* \frac{\partial V}{\partial x}\psi\, dx - \int_{-\infty}^{\infty} V\psi^* \frac{\partial \psi}{\partial x}\, dx$$

Canceling the equal and opposite terms $\int_{-\infty}^{\infty} V\psi^* \frac{\partial \psi}{\partial x}\, dx$ yields:

$$\frac{d\langle p\rangle}{dt} = -\int_{-\infty}^{\infty} \psi^* \frac{\partial V}{\partial x}\psi\, dx$$

Recognizing the right-hand side as the negative expectation value of the spatial gradient of potential energy, we obtain:

$$\frac{d\langle p\rangle}{dt} = -\left\langle \frac{\partial V}{\partial x} \right\rangle$$

This completes the proof of Ehrenfest's theorem.

3. Stationary States

Definition of Stationary States

A stationary state is a quantum mechanical state corresponding to a definite, well-defined energy E. For a system whose potential energy $V(x)$ is independent of time t, the Hamiltonian operator $\hat{H}$ is time-independent:

$$\hat{H} = -\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2} + V(x)$$

Under these conditions, the total wave function $\psi(x,t)$ satisfying the time-dependent Schrödinger equation can be separated into a product of a purely spatial function $\psi(x)$ and a time-dependent phase factor:

$$\psi(x,t) = \psi(x)\exp\left(-\frac{iEt}{\hbar}\right)$$

The spatial function $\psi(x)$ is an eigenfunction of the Hamiltonian operator, satisfying the time-independent Schrödinger equation:

$$\hat{H}\psi(x) = E\psi(x)$$

States described by such wave functions are called stationary states because all physical observables associated with the state remain constant over time.

Time-Independence of Probability Density

The probability density $P(x,t)$ of finding a particle at position x at time t in a stationary state is defined as:

$$P(x,t) = |\psi(x,t)|^2 = \psi^*(x,t)\psi(x,t)$$

Substituting the separable form $\psi(x,t) = \psi(x)\exp\left(-\frac{iEt}{\hbar}\right)$ and its complex conjugate $\psi^*(x,t) = \psi^*(x)\exp\left(\frac{iEt}{\hbar}\right)$ into the probability density gives:

$$P(x,t) = \left[\psi^*(x)\exp\left(\frac{iEt}{\hbar}\right)\right]\left[\psi(x)\exp\left(-\frac{iEt}{\hbar}\right)\right]$$

Rearranging the terms yields:

$$P(x,t) = \psi^*(x)\psi(x)\exp\left(\frac{iEt}{\hbar} - \frac{iEt}{\hbar}\right)$$

Simplifying the exponential factor yields $\exp(0) = 1$, giving:

$$P(x,t) = \psi^*(x)\psi(x) = |\psi(x)|^2$$

Since $|\psi(x)|^2$ contains no explicit dependence on time t, the probability density of a stationary state is strictly time-independent:

$$\frac{\partial P(x,t)}{\partial t} = 0$$

Expectation Values in Stationary States

Let $\hat{A}$ be any dynamical operator that does not depend explicitly on time t. The expectation value of $\hat{A}$ in a stationary state $\psi(x,t)$ is given by:

$$\langle A\rangle = \int_{-\infty}^{\infty} \psi^*(x,t)\hat{A}\psi(x,t)\,dx$$

Substituting the stationary state wave function $\psi(x,t) = \psi(x)\exp\left(-\frac{iEt}{\hbar}\right)$ and its complex conjugate yields:

$$\langle A\rangle = \int_{-\infty}^{\infty} \left[\psi^*(x)\exp\left(\frac{iEt}{\hbar}\right)\right]\hat{A}\left[\psi(x)\exp\left(-\frac{iEt}{\hbar}\right)\right]\,dx$$

Since $\hat{A}$ acts only on spatial coordinates x and contains no time derivatives or explicit time dependence, the time-dependent phase factor $\exp\left(-\frac{iEt}{\hbar}\right)$ passes through $\hat{A}$ unchanged:

$$\langle A\rangle = \int_{-\infty}^{\infty} \psi^*(x)\exp\left(\frac{iEt}{\hbar}\right)\exp\left(-\frac{iEt}{\hbar}\right)\hat{A}\psi(x)\,dx$$

Combining the complex conjugate exponential factors yields $\exp(0) = 1$:

$$\langle A \rangle = \int_{-\infty}^{\infty} \psi^*(x) \hat{A} \psi(x) \, dx$$

Because the resulting integrand is completely independent of time t, the expectation value of any time-independent operator in a stationary state is constant in time:

$$\frac{d\langle A \rangle}{dt} = 0$$

4. Generalised Uncertainty Principle

Derivation from Schwarz Inequality for Two Hermitian Operators

Let $\hat{A}$ and $\hat{B}$ be two Hermitian operators corresponding to physical observables. The expectation values of these operators in a normalized state $|\psi\rangle$ are denoted by $\langle \hat{A} \rangle = \langle \psi | \hat{A} | \psi \rangle$ and $\langle \hat{B} \rangle = \langle \psi | \hat{B} | \psi \rangle$. We define the deviation operators $\Delta\hat{A}$ and $\Delta\hat{B}$ as:

$$\Delta\hat{A} = \hat{A} - \langle \hat{A} \rangle \hat{I}$$

$$\Delta\hat{B} = \hat{B} - \langle \hat{B} \rangle \hat{I}$$

Since $\hat{A}$ and $\hat{B}$ are Hermitian and their expectation values $\langle \hat{A} \rangle$ and $\langle \hat{B} \rangle$ are real numbers, the deviation operators $\Delta\hat{A}$ and $\Delta\hat{B}$ are also Hermitian. We construct two state vectors $|\psi_1\rangle$ and $|\psi_2\rangle$ by applying these deviation operators to the state $|\psi\rangle$:

$$|\psi_1\rangle = \Delta\hat{A}|\psi\rangle$$

$$|\psi_2\rangle = \Delta\hat{B}|\psi\rangle$$

The inner products of these vectors with themselves yield the variances $(\Delta A)^2$ and $(\Delta B)^2$:

$$\langle \psi_1 | \psi_1 \rangle = \langle \psi | (\Delta\hat{A})^\dagger \Delta\hat{A} | \psi \rangle = \langle \psi | (\Delta\hat{A})^2 | \psi \rangle = (\Delta A)^2$$

$$\langle \psi_2 | \psi_2 \rangle = \langle \psi | (\Delta\hat{B})^\dagger \Delta\hat{B} | \psi \rangle = \langle \psi | (\Delta\hat{B})^2 | \psi \rangle = (\Delta B)^2$$

According to the Schwarz inequality for any two vectors in Hilbert space:

$$\langle \psi_1 | \psi_1 \rangle \langle \psi_2 | \psi_2 \rangle \geq |\langle \psi_1 | \psi_2 \rangle|^2$$

Substituting the norms into the Schwarz inequality gives:

$$(\Delta A)^2 (\Delta B)^2 \geq |\langle \Delta\hat{A}\Delta\hat{B} \rangle|^2$$

We decompose the product of the deviation operators into symmetric and anti-symmetric parts:

$$\Delta\hat{A}\Delta\hat{B} = \frac{1}{2}(\Delta\hat{A}\Delta\hat{B} - \Delta\hat{B}\Delta\hat{A}) + \frac{1}{2}(\Delta\hat{A}\Delta\hat{B} + \Delta\hat{B}\Delta\hat{A})$$

Using the commutator $[\Delta\hat{A}, \Delta\hat{B}] = \Delta\hat{A}\Delta\hat{B} - \Delta\hat{B}\Delta\hat{A}$ and the anti-commutator $\{\Delta\hat{A}, \Delta\hat{B}\} = \Delta\hat{A}\Delta\hat{B} + \Delta\hat{B}\Delta\hat{A}$, this identity becomes:

$$\Delta\hat{A}\Delta\hat{B} = \frac{1}{2}[\Delta\hat{A}, \Delta\hat{B}] + \frac{1}{2}\{\Delta\hat{A}, \Delta\hat{B}\}$$

The commutator of the deviation operators is identical to the commutator of the original operators:

$$[\Delta\hat{A}, \Delta\hat{B}] = [\hat{A} - \langle\hat{A}\rangle, \hat{B} - \langle\hat{B}\rangle] = [\hat{A}, \hat{B}]$$

Taking the expectation value of both sides gives:

$$\langle\Delta\hat{A}\Delta\hat{B}\rangle = \frac{1}{2}\langle[\hat{A}, \hat{B}]\rangle + \frac{1}{2}\langle\{\Delta\hat{A}, \Delta\hat{B}\}\rangle$$

Since $\hat{A}$ and $\hat{B}$ are Hermitian, their commutator is anti-Hermitian:

$$[\hat{A}, \hat{B}]^\dagger = (\hat{A}\hat{B} - \hat{B}\hat{A})^\dagger = \hat{B}^\dagger\hat{A}^\dagger - \hat{A}^\dagger\hat{B}^\dagger = \hat{B}\hat{A} - \hat{A}\hat{B} = -[\hat{A}, \hat{B}]$$

Therefore, its expectation value $\langle[\hat{A}, \hat{B}]\rangle$ is purely imaginary. On the other hand, the anti-commutator is Hermitian:

$$\{\Delta\hat{A}, \Delta\hat{B}\}^\dagger = (\Delta\hat{A}\Delta\hat{B} + \Delta\hat{B}\Delta\hat{A})^\dagger = \Delta\hat{B}\Delta\hat{A} + \Delta\hat{A}\Delta\hat{B} = \{\Delta\hat{A}, \Delta\hat{B}\}$$

So its expectation value $\langle\{\Delta\hat{A}, \Delta\hat{B}\}\rangle$ is purely real. For a complex number composed of independent real and imaginary parts, the square of the absolute value is the sum of the squares of these parts:

$$|\langle\Delta\hat{A}\Delta\hat{B}\rangle|^2 = \frac{1}{4}|\langle[\hat{A}, \hat{B}]\rangle|^2 + \frac{1}{4}\langle\{\Delta\hat{A}, \Delta\hat{B}\}\rangle^2$$

Substituting this back into the Schwarz inequality yields the full generalized uncertainty principle:

$$(\Delta A)^2(\Delta B)^2 \geq \frac{1}{4}|\langle[\hat{A}, \hat{B}]\rangle|^2 + \frac{1}{4}\langle\{\Delta\hat{A}, \Delta\hat{B}\}\rangle^2$$

Since the anti-commutator term $\frac{1}{4}\langle\{\Delta\hat{A}, \Delta\hat{B}\}\rangle^2$ is non-negative, dropping it gives the standard Heisenberg form of the uncertainty principle:

$$(\Delta A)^2(\Delta B)^2 \geq \frac{1}{4}|\langle[\hat{A}, \hat{B}]\rangle|^2$$

Taking the positive square root on both sides produces:

$$\Delta A\Delta B \geq \frac{1}{2}|\langle[\hat{A}, \hat{B}]\rangle|$$

Heisenberg Uncertainty Relation for Position and Momentum

For position operator $\hat{x}$ and momentum operator $\hat{p}$, the fundamental commutation relation is:

$$[\hat{x}, \hat{p}] = i\hbar$$

Taking the expectation value in any state gives:

$$\langle [\hat{x}, \hat{p}] \rangle = i\hbar$$

The absolute magnitude of this expectation value is:

$$|\langle [\hat{x}, \hat{p}] \rangle| = |i\hbar| = \hbar$$

Substituting this value into the Heisenberg form of the generalized uncertainty principle yields:

$$\Delta x \Delta p \geq \frac{1}{2}\hbar$$

$$\Delta x \Delta p \geq \frac{\hbar}{2}$$

Verification for Ground State of One-Dimensional Harmonic Oscillator

The normalized ground-state wave function of a one-dimensional harmonic oscillator of mass m and frequency ω is:

$$\psi_0(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} \exp\left(-\frac{m\omega}{2\hbar}x^2\right)$$

The expectation value of position $\langle x \rangle$ is:

$$\langle x \rangle = \int_{-\infty}^{\infty} \psi_0^*(x) x \psi_0(x) dx = \left(\frac{m\omega}{\pi\hbar}\right)^{1/2} \int_{-\infty}^{\infty} x \exp\left(-\frac{m\omega}{\hbar}x^2\right) dx$$

Since the integrand is an odd function over a symmetric interval, the integral vanishes:

$$\langle x \rangle = 0$$

The expectation value of x^2 is:

$$\langle x^2 \rangle = \int_{-\infty}^{\infty} \psi_0^*(x) x^2 \psi_0(x) dx = \left(\frac{m\omega}{\pi\hbar}\right)^{1/2} \int_{-\infty}^{\infty} x^2 \exp\left(-\frac{m\omega}{\hbar}x^2\right) dx$$

Using the Gaussian integral identity $\int_{-\infty}^{\infty} x^2 e^{-\alpha x^2} dx = \frac{1}{2\alpha}\sqrt{\frac{\pi}{\alpha}}$ with $\alpha = \frac{m\omega}{\hbar}$:

$$\int_{-\infty}^{\infty} x^2 \exp\left(-\frac{m\omega}{\hbar}x^2\right) dx = \frac{\hbar}{2m\omega}\sqrt{\frac{\pi\hbar}{m\omega}}$$

Substituting this back gives:

$$\langle x^2 \rangle = \left(\frac{m\omega}{\pi\hbar}\right)^{1/2} \left(\frac{\hbar}{2m\omega}\sqrt{\frac{\pi\hbar}{m\omega}}\right) = \frac{\hbar}{2m\omega}$$

The position uncertainty Δx is therefore:

$$\Delta x = \sqrt{\langle x^2 \rangle - \langle x \rangle^2} = \sqrt{\frac{\hbar}{2m\omega} - 0} = \sqrt{\frac{\hbar}{2m\omega}}$$

For momentum, the expectation value $\langle p \rangle$ is:

$$\langle p \rangle = \int_{-\infty}^{\infty} \psi_0^*(x) \left(-i\hbar \frac{d}{dx} \right) \psi_0(x) dx$$

Differentiating the ground-state wave function yields:

$$\frac{d}{dx}\psi_0(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} \left(-\frac{m\omega}{\hbar}x\right) \exp\left(-\frac{m\omega}{2\hbar}x^2\right)$$

Substituting this derivative into the integral gives an odd integrand, which vanishes:

$$\langle p \rangle = 0$$

The expectation value of p^2 is:

$$\langle p^2 \rangle = \int_{-\infty}^{\infty} \psi_0^*(x) \left(-\hbar^2 \frac{d^2}{dx^2} \right) \psi_0(x) dx$$

Evaluating the second derivative of $\psi_0(x)$:

$$\frac{d^2}{dx^2}\psi_0(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} \left[-\frac{m\omega}{\hbar} + \left(\frac{m\omega}{\hbar}\right)^2 x^2\right] \exp\left(-\frac{m\omega}{2\hbar}x^2\right)$$

Substituting this derivative into the integral for $\langle p^2 \rangle$:

$$\langle p^2 \rangle = -\hbar^2 \left(\frac{m\omega}{\pi\hbar}\right)^{1/2} \int_{-\infty}^{\infty} \left[-\frac{m\omega}{\hbar} + \left(\frac{m\omega}{\hbar}\right)^2 x^2\right] \exp\left(-\frac{m\omega}{\hbar}x^2\right) dx$$

Evaluating the standard Gaussian integrals with $\alpha = \frac{m\omega}{\hbar}$:

$$\langle p^2 \rangle = -\hbar^2 \left(\frac{m\omega}{\pi\hbar}\right)^{1/2} \left[-\frac{m\omega}{\hbar}\sqrt{\frac{\pi\hbar}{m\omega}} + \left(\frac{m\omega}{\hbar}\right)^2 \left(\frac{\hbar}{2m\omega}\sqrt{\frac{\pi\hbar}{m\omega}}\right)\right]$$

$$\langle p^2 \rangle = -\hbar^2 \left[-\frac{m\omega}{\hbar} + \frac{m\omega}{2\hbar}\right] = -\hbar^2 \left(-\frac{m\omega}{2\hbar}\right) = \frac{m\hbar\omega}{2}$$

The momentum uncertainty Δp is therefore:

$$\Delta p = \sqrt{\langle p^2 \rangle - \langle p \rangle^2} = \sqrt{\frac{m\hbar\omega}{2} - 0} = \sqrt{\frac{m\hbar\omega}{2}}$$

Multiplying the uncertainties Δx and Δp together gives:

$$\Delta x \Delta p = \sqrt{\frac{\hbar}{2m\omega}}\sqrt{\frac{m\hbar\omega}{2}} = \frac{\hbar}{2}$$

Thus, the product of uncertainties for the ground state of the harmonic oscillator is exactly equal to $\frac{\hbar}{2}$.

Estimation of Zero-Point Energy of Harmonic Oscillator

The classical energy expression for a harmonic oscillator of mass m and angular frequency ω is:

$$E = \frac{p^2}{2m} + \frac{1}{2}m\omega^2 x^2$$

For a state symmetric about the origin, $\langle x \rangle = 0$ and $\langle p \rangle = 0$, so $\langle x^2 \rangle = (\Delta x)^2$ and $\langle p^2 \rangle = (\Delta p)^2$. The expectation value of energy is written in terms of these uncertainties as:

$$E = \frac{(\Delta p)^2}{2m} + \frac{1}{2}m\omega^2 (\Delta x)^2$$

Using the Heisenberg uncertainty relation at its limit $\Delta p \approx \frac{\hbar}{2\Delta x}$, we substitute for Δp:

$$E(\Delta x) = \frac{1}{2m}\left(\frac{\hbar}{2\Delta x}\right)^2 + \frac{1}{2}m\omega^2(\Delta x)^2$$

$$E(\Delta x) = \frac{\hbar^2}{8m(\Delta x)^2} + \frac{1}{2}m\omega^2(\Delta x)^2$$

To estimate the ground-state energy, we minimize $E(\Delta x)$ with respect to Δx by setting its derivative to zero:

$$\frac{dE}{d(\Delta x)} = -\frac{\hbar^2}{4m(\Delta x)^3} + m\omega^2 \Delta x = 0$$

Rearranging this derivative equation yields:

$$m\omega^2 \Delta x = \frac{\hbar^2}{4m(\Delta x)^3}$$

$$(\Delta x)^4 = \frac{\hbar^2}{4m^2\omega^2}$$

Taking the square root gives the position uncertainty corresponding to minimum energy:

$$(\Delta x)^2 = \frac{\hbar}{2m\omega}$$

Substituting $(\Delta x)^2 = \frac{\hbar}{2m\omega}$ back into the energy expression gives the zero-point energy E_0:

$$E_0 = \frac{\hbar^2}{8m\left(\frac{\hbar}{2m\omega}\right)} + \frac{1}{2}m\omega^2\left(\frac{\hbar}{2m\omega}\right)$$

$$E_0 = \frac{1}{4}\hbar\omega + \frac{1}{4}\hbar\omega = \frac{1}{2}\hbar\omega$$

This confirms that the zero-point energy of the harmonic oscillator estimated from the uncertainty principle is $E_0 = \frac{1}{2}\hbar\omega$.

5. Matrix Representation of Operators

Expansion of Wave Function in Complete Orthonormal Set

Let $\{\phi_n(x)\}$ be a complete set of orthonormal basis functions defined over a given domain, satisfying the orthonormality relation:

$$\int \phi_m^*(x)\phi_n(x)dx = \delta_{mn}$$

where δ_{mn} is the Kronecker delta defined as $\delta_{mn} = 1$ for $m = n$ and $\delta_{mn} = 0$ for $m \neq n$. Because the set $\{\phi_n(x)\}$ is complete, any arbitrary square-integrable wave function $\psi(x)$ can be expanded as a linear combination of these basis functions:

$$\psi(x) = \sum_n c_n\phi_n(x)$$

To determine the expansion coefficients c_n, we multiply both sides of this expansion by $\phi_m^*(x)$ and integrate over all space:

$$\int \phi_m^*(x)\psi(x)dx = \int \phi_m^*(x)\left(\sum_n c_n\phi_n(x)\right)dx$$

Interchanging the summation and integration operations yields:

$$\int \phi_m^*(x)\psi(x)dx = \sum_n c_n \int \phi_m^*(x)\phi_n(x)dx$$

Applying the orthonormality condition $\int \phi_m^*(x)\phi_n(x)dx = \delta_{mn}$ reduces the right-hand side to:

$$\int \phi_m^*(x)\psi(x)dx = \sum_n c_n\delta_{mn} = c_m$$

Thus, the expansion coefficients c_n are given by:

$$c_n = \int \phi_n^*(x)\psi(x)dx$$

In matrix representation, the state wave function $\psi(x)$ is completely specified by the column vector C formed by these expansion coefficients:

$$C = \begin{pmatrix} c_1 \\ c_2 \\ c_3 \\ \vdots \end{pmatrix}$$

Matrix Elements of an Operator and Matrix Multiplication

Consider an operator $\hat{A}$ acting on a basis function $\phi_n(x)$. Since the set $\{\phi_m(x)\}$ is complete, the resulting function $\hat{A}\phi_n(x)$ can itself be expanded in the same basis set:

$$\hat{A}\phi_n(x) = \sum_m A_{mn}\phi_m(x)$$

To isolate the coefficient A_{mn}, we multiply both sides by $\phi_m^*(x)$ and integrate over all space:

$$\int \phi_m^*(x)\hat{A}\phi_n(x)dx = \int \phi_m^*(x)\left(\sum_k A_{kn}\phi_k(x)\right)dx$$

Interchanging the summation and integration gives:

$$\int \phi_m^*(x)\hat{A}\phi_n(x)dx = \sum_k A_{kn}\int \phi_m^*(x)\phi_k(x)dx = \sum_k A_{kn}\delta_{mk} = A_{mn}$$

Thus, the matrix elements A_{mn} of the operator $\hat{A}$ in the basis $\{\phi_n(x)\}$ are defined as:

$$A_{mn} = \int \phi_m^*(x)\hat{A}\phi_n(x)dx = \langle\phi_m|\hat{A}|\phi_n\rangle$$

These elements form a matrix A that represents the operator $\hat{A}$:

$$A = \begin{pmatrix} A_{11} & A_{12} & A_{13} & \dots \\ A_{21} & A_{22} & A_{23} & \dots \\ A_{31} & A_{32} & A_{33} & \dots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}$$

Now, consider the action of the operator $\hat{A}$ on an arbitrary state $\psi(x)$ to produce a new state $\phi(x)$:

$$\phi(x) = \hat{A}\psi(x)$$

We expand both $\psi(x)$ and $\phi(x)$ in the orthonormal basis $\{\phi_n(x)\}$:

$$\psi(x) = \sum_n c_n\phi_n(x)$$

$$\phi(x) = \sum_m d_m\phi_m(x)$$

Substituting the expansion of $\psi(x)$ into $\phi(x) = \hat{A}\psi(x)$ yields:

$$\sum_m d_m \phi_m(x) = \hat{A}\left(\sum_n c_n \phi_n(x)\right) = \sum_n c_n \hat{A}\phi_n(x)$$

Substituting $\hat{A}\phi_n(x) = \sum_m A_{mn}\phi_m(x)$ into the right-hand side gives:

$$\sum_m d_m \phi_m(x) = \sum_n c_n \left(\sum_m A_{mn}\phi_m(x)\right) = \sum_m \left(\sum_n A_{mn} c_n\right)\phi_m(x)$$

Equating coefficients of $\phi_m(x)$ on both sides yields:

$$d_m = \sum_n A_{mn} c_n$$

In matrix notation, letting D be the column vector of coefficients d_m and C be the column vector of coefficients c_n, this relation is written as matrix multiplication:

$$D = AC$$

$$\begin{pmatrix} d_1 \\ d_2 \\ \vdots \end{pmatrix} = \begin{pmatrix} A_{11} & A_{12} & \cdots \\ A_{21} & A_{22} & \cdots \\ \vdots & \vdots & \ddots \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ \vdots \end{pmatrix}$$

Hence, the operation of a quantum mechanical operator on a wave function translates directly to matrix multiplication of its matrix representation with the column vector of the wave function.

Normalization and Orthogonality in Matrix Form

Let $\psi(x) = \sum_n c_n \phi_n(x)$ and $\chi(x) = \sum_n d_n \phi_n(x)$ be two wave functions expanded in the orthonormal basis $\{\phi_n(x)\}$. The inner product of $\chi(x)$ and $\psi(x)$ is:

$$\langle \chi | \psi \rangle = \int \chi^*(x)\psi(x)dx = \int \left(\sum_m d_m \phi_m(x)\right)^* \left(\sum_n c_n \phi_n(x)\right) dx$$

Expanding the complex conjugation and moving the integrals inside the sums gives:

$$\langle \chi | \psi \rangle = \sum_m \sum_n d_m^* c_n \int \phi_m^*(x)\phi_n(x)dx$$

Using the orthonormality condition $\int \phi_m^*(x)\phi_n(x)dx = \delta_{mn}$, this reduces to:

$$\langle \chi | \psi \rangle = \sum_m \sum_n d_m^* c_n \delta_{mn} = \sum_n d_n^* c_n$$

In matrix notation, defining the conjugate transpose (Hermitian conjugate) row vector $D^\dagger$ of the column vector D:

$$D^\dagger = \begin{pmatrix} d_1^* & d_2^* & d_3^* & \dots \end{pmatrix}$$

the inner product is expressed as:

$$\langle \chi | \psi \rangle = D^\dagger C = \begin{pmatrix} d_1^* & d_2^* & \dots \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ \vdots \end{pmatrix} = \sum_n d_n^* c_n$$

For a single normalized wave function $\psi(x)$, setting $\chi(x) = \psi(x)$ (which implies $d_n = c_n$) gives:

$$\langle \psi | \psi \rangle = \int |\psi(x)|^2 dx = \sum_n c_n^* c_n = \sum_n |c_n|^2 = 1$$

In matrix form, the normalization condition is written as:

$$C^\dagger C = 1$$

$$\begin{pmatrix} c_1^* & c_2^* & c_3^* & \dots \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ c_3 \\ \vdots \end{pmatrix} = \sum_n |c_n|^2 = 1$$

Furthermore, the orthonormality of the basis functions $\{\phi_n(x)\}$ is represented in matrix form by the matrix elements of the identity operator $\hat{I}$:

$$I_{mn} = \langle \phi_m | \hat{I} | \phi_n \rangle = \int \phi_m^*(x) \phi_n(x) dx = \delta_{mn}$$

This yields the unit matrix (identity matrix) I:

$$I = \begin{pmatrix} 1 & 0 & 0 & \dots \\ 0 & 1 & 0 & \dots \\ 0 & 0 & 1 & \dots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}$$

This establishes the normalization and orthonormality of wave functions in matrix representation.

6. Coordinate and Momentum Representations

Wave Functions in Coordinate and Momentum Representations

In quantum mechanics, a physical state $|\psi\rangle$ can be described in different representations depending on the choice of basis vectors used to span the state space. The two most fundamental representations are the

coordinate (or position) representation and the momentum representation.

In the coordinate representation, the basis consists of the continuous set of position eigenstates $|x\rangle$ satisfying the completeness relation:

$$\int_{-\infty}^{\infty} |x\rangle\langle x| dx = I$$

The wave function in coordinate representation, denoted by $\psi(x)$, is defined as the projection of the state vector $|\psi\rangle$ onto the position eigenstate $|x\rangle$:

$$\psi(x) = \langle x|\psi\rangle$$

Similarly, in the momentum representation, the basis consists of the continuous set of momentum eigenstates $|p\rangle$ satisfying the completeness relation:

$$\int_{-\infty}^{\infty} |p\rangle\langle p| dp = I$$

The wave function in momentum representation, denoted by $\phi(p)$, is defined as the projection of the state vector $|\psi\rangle$ onto the momentum eigenstate $|p\rangle$:

$$\phi(p) = \langle p|\psi\rangle$$

The transformation function connecting the coordinate and momentum basis states is given by the free-particle plane wave state:

$$\langle x|p\rangle = \frac{1}{\sqrt{2\pi\hbar}} e^{ipx/\hbar}$$

Taking the complex conjugate of this scalar product gives the reverse transformation function:

$$\langle p|x\rangle = \langle x|p\rangle^* = \frac{1}{\sqrt{2\pi\hbar}} e^{-ipx/\hbar}$$

To obtain the momentum wave function $\phi(p)$ from the coordinate wave function $\psi(x)$, we insert the completeness relation for the position basis into the definition of $\phi(p)$:

$$\phi(p) = \langle p|\psi\rangle$$

Inserting the identity operator expressed in the position basis gives:

$$\phi(p) = \langle p| \left(\int_{-\infty}^{\infty} |x\rangle\langle x| dx \right) |\psi\rangle$$

Moving the integration outside the inner product yields:

$$\phi(p) = \int_{-\infty}^{\infty} \langle p|x\rangle\langle x|\psi\rangle dx$$

Substituting $\langle x|\psi\rangle = \psi(x)$ into the integrand gives:

$$\phi(p) = \int_{-\infty}^{\infty} \langle p|x\rangle \psi(x) dx$$

Substituting the explicit form of $\langle p|x\rangle = \frac{1}{\sqrt{2\pi\hbar}} e^{-ipx/\hbar}$ yields:

$$\phi(p) = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\hbar}} e^{-ipx/\hbar} \psi(x) dx$$

Rearranging terms leads to the forward Fourier transform connecting $\psi(x)$ to $\phi(p)$:

$$\phi(p) = \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{\infty} \psi(x) e^{-ipx/\hbar} dx$$

Conversely, to express the coordinate wave function $\psi(x)$ in terms of the momentum wave function $\phi(p)$, we start from the definition of $\psi(x)$:

$$\psi(x) = \langle x|\psi\rangle$$

Inserting the completeness relation for the momentum basis gives:

$$\psi(x) = \langle x| \left(\int_{-\infty}^{\infty} |p\rangle\langle p| dp \right) |\psi\rangle$$

Moving the integration outside the inner product yields:

$$\psi(x) = \int_{-\infty}^{\infty} \langle x|p\rangle \langle p|\psi\rangle dp$$

Substituting $\langle p|\psi\rangle = \phi(p)$ into the integrand gives:

$$\psi(x) = \int_{-\infty}^{\infty} \langle x|p\rangle \phi(p) dp$$

Substituting the explicit form of $\langle x|p\rangle = \frac{1}{\sqrt{2\pi\hbar}} e^{ipx/\hbar}$ yields:

$$\psi(x) = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\hbar}} e^{ipx/\hbar} \phi(p) dp$$

Rearranging terms leads to the inverse Fourier transform:

$$\psi(x) = \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{\infty} \phi(p) e^{ipx/\hbar} dp$$

Schrödinger Wave Equation in Momentum Representation

To formulate the Schrödinger wave equation in momentum space, we must determine how the dynamical operators act on the momentum wave function $\phi(p)$.

In momentum representation, the momentum operator $\hat{p}$ acting on $\phi(p)$ corresponds simply to multiplication by the continuous variable p:

$$\hat{p}\phi(p) = p\phi(p)$$

To find the form of the position operator $\hat{x}$ in momentum representation, we evaluate its matrix element $\langle p|\hat{x}|\psi\rangle$. Inserting the identity operator in coordinate representation gives:

$$\langle p|\hat{x}|\psi\rangle = \langle p|\hat{x}\left(\int_{-\infty}^{\infty}|x\rangle\langle x|dx\right)|\psi\rangle$$

Pulling the integration outside yields:

$$\langle p|\hat{x}|\psi\rangle = \int_{-\infty}^{\infty}\langle p|\hat{x}|x\rangle\langle x|\psi\rangle dx$$

Since $|x\rangle$ is an eigenstate of the position operator with eigenvalue x, we have $\hat{x}|x\rangle = x|x\rangle$:

$$\langle p|\hat{x}|\psi\rangle = \int_{-\infty}^{\infty}x\langle p|x\rangle\langle x|\psi\rangle dx$$

Substituting $\langle x|\psi\rangle = \psi(x)$ and $\langle p|x\rangle = \frac{1}{\sqrt{2\pi\hbar}}e^{-ipx/\hbar}$ gives:

$$\langle p|\hat{x}|\psi\rangle = \int_{-\infty}^{\infty}x\left(\frac{1}{\sqrt{2\pi\hbar}}e^{-ipx/\hbar}\right)\psi(x)dx$$

We note the differentiation identity with respect to momentum p:

$$\frac{d}{dp}e^{-ipx/\hbar} = -\frac{ix}{\hbar}e^{-ipx/\hbar}$$

Multiplying both sides by $i\hbar$ gives:

$$i\hbar\frac{d}{dp}e^{-ipx/\hbar} = xe^{-ipx/\hbar}$$

Substituting this identity into the integral expression gives:

$$\langle p|\hat{x}|\psi\rangle = \int_{-\infty}^{\infty}\left(i\hbar\frac{d}{dp}\frac{1}{\sqrt{2\pi\hbar}}e^{-ipx/\hbar}\right)\psi(x)dx$$

Interchanging the order of differentiation with respect to p and integration over x yields:

$$\langle p|\hat{x}|\psi\rangle = i\hbar\frac{d}{dp}\left(\frac{1}{\sqrt{2\pi\hbar}}\int_{-\infty}^{\infty}\psi(x)e^{-ipx/\hbar}dx\right)$$

Recognizing the term in parentheses as the Fourier transform defining $\phi(p)$, we obtain:

$$\langle p|\hat{x}|\psi\rangle = i\hbar\frac{d}{dp}\phi(p)$$

Thus, in the momentum representation, the position operator is represented by the differential operator:

$$\hat{x} = i\hbar\frac{d}{dp}$$

The time-independent Schrödinger equation in abstract state space is:

$$\hat{H}|\psi\rangle = E|\psi\rangle$$

Expressing the Hamiltonian operator as $\hat{H} = \frac{\hat{p}^2}{2m} + V(\hat{x})$, the eigenvalue equation becomes:

$$\left(\frac{\hat{p}^2}{2m} + V(\hat{x})\right)|\psi\rangle = E|\psi\rangle$$

Projecting this equation onto the momentum eigenstate bra $\langle p|$ gives:

$$\langle p|\left(\frac{\hat{p}^2}{2m} + V(\hat{x})\right)|\psi\rangle = E\langle p|\psi\rangle$$

Distributing the bra state $\langle p|$ across the operator terms yields:

$$\frac{1}{2m}\langle p|\hat{p}^2|\psi\rangle + \langle p|V(\hat{x})|\psi\rangle = E\phi(p)$$

Since $\langle p|\hat{p} = p\langle p|$, the kinetic energy term simplifies to:

$$\frac{p^2}{2m}\langle p|\psi\rangle + \langle p|V(\hat{x})|\psi\rangle = E\phi(p)$$

Substituting $\langle p|\psi\rangle = \phi(p)$ gives:

$$\frac{p^2}{2m}\phi(p) + \langle p|V(\hat{x})|\psi\rangle = E\phi(p)$$

Replacing the coordinate operator $\hat{x}$ by $i\hbar\frac{d}{dp}$ in the potential energy function gives:

$$\frac{p^2}{2m}\phi(p) + V\left(i\hbar\frac{d}{dp}\right)\phi(p) = E\phi(p)$$

Factoring out the momentum wave function $\phi(p)$ gives the time-independent Schrödinger wave equation in momentum representation:

$$\left(\frac{p^2}{2m} + V\left(i\hbar\frac{d}{dp}\right)\right)\phi(p) = E\phi(p)$$

Momentum Eigenfunction

An eigenfunction of the momentum operator corresponding to a definite momentum eigenvalue p_0 must satisfy the momentum eigenvalue equation in momentum space:

$$\hat{p}\phi_{p_0}(p) = p_0\phi_{p_0}(p)$$

Since the momentum operator $\hat{p}$ acts simply as multiplication by p in momentum representation, this equation takes the form:

$$p\phi_{p_0}(p) = p_0\phi_{p_0}(p)$$

This equation implies that $\phi_{p_0}(p)$ must vanish everywhere except at $p = p_0$. The normalized eigenfunction in momentum representation is therefore given by the Dirac delta function:

$$\phi_{p_0}(p) = \delta(p - p_0)$$

To find the corresponding momentum eigenfunction in the coordinate representation, we use the inverse Fourier transform:

$$\psi_{p_0}(x) = \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{\infty} \phi_{p_0}(p) e^{ipx/\hbar} dp$$

Substituting $\phi_{p_0}(p) = \delta(p - p_0)$ into the integral yields:

$$\psi_{p_0}(x) = \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{\infty} \delta(p - p_0) e^{ipx/\hbar} dp$$

Applying the sifting property of the Dirac delta function $\int_{-\infty}^{\infty} \delta(p - p_0) f(p) dp = f(p_0)$ gives the momentum eigenfunction in coordinate representation:

$$\psi_{p_0}(x) = \frac{1}{\sqrt{2\pi\hbar}} e^{ip_0x/\hbar}$$

Significance of Momentum Wave Function

The momentum wave function $\phi(p)$ provides a complete probabilistic description of a physical system in momentum space, completely equivalent to the coordinate wave function $\psi(x)$ in position space.

The primary physical interpretation of $\phi(p)$ is established through Born's statistical interpretation. The absolute square of the momentum wave function, $|\phi(p)|^2$, represents the probability density in momentum space. Specifically, the probability $dP(p)$ of finding the particle's momentum in the infinitesimal range between p and $p + dp$ is given by:

$$dP(p) = |\phi(p)|^2 dp$$

For a normalized physical state, the total probability of finding the particle with some momentum across the entire momentum domain must be equal to unity:

$$\int_{-\infty}^{\infty} |\phi(p)|^2 dp = 1$$

The normalization of the momentum wave function is directly preserved from coordinate space via Parseval's theorem for Fourier transforms:

$$\int_{-\infty}^{\infty} |\psi(x)|^2 dx = \int_{-\infty}^{\infty} |\phi(p)|^2 dp = 1$$

In the case of a localized wave packet, the momentum wave function reveals the underlying distribution of linear momentum components composing the state. A wave packet that is tightly confined in coordinate space (Δx is small) requires a broad range of harmonic plane waves to construct through superposition. Consequently, its momentum wave function $\phi(p)$ is widely spread out (Δp is large). Conversely, if the wave packet is broadly spread in position space (Δx is large), its momentum wave function is sharply peaked around a central momentum value p_0 (Δp is small). This inverse relationship between the spatial extent and the momentum spread of a wave packet is a direct mathematical consequence of the Fourier transform pair connecting $\psi(x)$ and $\phi(p)$, providing the physical foundation for the uncertainty principle $\Delta x \Delta p \geq \frac{\hbar}{2}$.

7. Dirac's Bra and Ket Notation

Dirac introduced a concise and powerful symbolic notation that abstracts the state of a physical system from specific representations such as position wave functions or column matrices. In this formalism, physical states and physical quantities are represented by elements of a complex vector space known as Hilbert space and linear operators acting upon it.

State Vectors: Kets and Bras

A quantum state of a physical system is represented by a state vector called a **ket**, denoted by the symbol $|\psi\rangle$. The label ψ inside the ket identifies the particular physical state. Kets are elements of a complex vector space (Hilbert space $\mathcal{H}$).

To every ket vector $|\psi\rangle$ in the Hilbert space, there corresponds a dual vector in the dual space $\mathcal{H}^*$, called a **bra**, denoted by $\langle\psi|$. The bra vector $\langle\psi|$ is the Hermitian conjugate (or adjoint) of the ket vector $|\psi\rangle$. Mathematically, this association is expressed as:

$$(|\psi\rangle)^\dagger = \langle\psi|$$

If a state vector is multiplied by a complex scalar c, its dual vector undergoes complex conjugation:

$$(c|\psi\rangle)^\dagger = c^*\langle\psi|$$

Algebraic Operations on Kets and Bras

Linear operations in quantum mechanics satisfy standard vector space properties for both kets and bras.

1. Addition of Kets and Bras

If $|\psi_1\rangle$ and $|\psi_2\rangle$ are two state vectors in Hilbert space, their sum is another ket $|\psi_3\rangle$ in the same space:

$$|\psi_3\rangle = |\psi_1\rangle + |\psi_2\rangle$$

Taking the Hermitian conjugate of this vector equation yields the corresponding relation for dual bra vectors:

$$\langle\psi_3| = \langle\psi_1| + \langle\psi_2|$$

2. Multiplication by a Complex Scalar

Multiplying a ket $|\psi\rangle$ by a complex number a yields a new ket vector:

$$|\phi\rangle = a|\psi\rangle$$

Taking the Hermitian conjugate of both sides gives the corresponding bra vector:

$$\langle\phi| = a^*\langle\psi|$$

Inner Product and Outer Product

Inner Product

The inner product of a bra vector $\langle\phi|$ and a ket vector $|\psi\rangle$ is written by joining the bra and ket together as $\langle\phi|\psi\rangle$. The inner product maps two state vectors to a single complex number:

$$\langle\phi|\psi\rangle \in \mathbb{C}$$

The inner product satisfies the property of skew-symmetry under Hermitian conjugation:

$$\langle\phi|\psi\rangle = \langle\psi|\phi\rangle^*$$

For any physically acceptable state vector $|\psi\rangle$, the inner product of the ket with its corresponding bra gives the square of the norm of the vector, which is a real and non-negative quantity:

$$\langle\psi|\psi\rangle \geq 0$$

If the state vector is normalized, its norm equals unity:

$$\langle\psi|\psi\rangle = 1$$

Outer Product

While the inner product $\langle\phi|\psi\rangle$ yields a complex scalar, placing a ket to the left of a bra forms an **outer product**, written as $|\psi\rangle\langle\phi|$. The outer product is a linear operator that acts on vectors in Hilbert space.

When the operator $|\psi\rangle\langle\phi|$ acts on an arbitrary ket vector $|\chi\rangle$, the associative property yields:

$$(|\psi\rangle\langle\phi|)|\chi\rangle = |\psi\rangle\langle\phi|\chi\rangle$$

Since $\langle\phi|\chi\rangle$ is a complex scalar $c = \langle\phi|\chi\rangle$, the result of this operation is a new ket proportional to $|\psi\rangle$:

$$(|\psi\rangle\langle\phi|)|\chi\rangle = c|\psi\rangle$$

Adjoint of an Operator

An operator $\hat{A}$ maps a ket vector $|\psi\rangle$ into another ket vector $|\psi'\rangle$:

$$|\psi'\rangle = \hat{A}|\psi\rangle$$

The adjoint (or Hermitian conjugate) of $\hat{A}$, denoted by $\hat{A}^\dagger$, is defined through the inner product relation for any arbitrary state vectors $|\phi\rangle$ and $|\psi\rangle$:

$$\langle\phi|\hat{A}^\dagger|\psi\rangle = \langle\psi|\hat{A}|\phi\rangle^*$$

The action of an operator $\hat{A}$ to the left on a bra vector $\langle\phi|$ produces a new bra vector:

$$\langle\phi'| = \langle\phi|\hat{A}$$

Taking the Hermitian conjugate of the state $|\psi'\rangle = \hat{A}|\psi\rangle$ gives:

$$(\hat{A}|\psi\rangle)^\dagger = \langle\psi|\hat{A}^\dagger$$

Orthonormality and Completeness Relation

Consider a set of discrete basis kets $\{|n\rangle\}$ spanning the Hilbert space.

Orthonormality

The basis kets are orthogonal to one another and individually normalized. This condition is expressed using the Kronecker delta δ_{mn}:

$$\langle m|n\rangle = \delta_{mn}$$

where:

$$\delta_{mn} = \begin{cases} 1 & \text{if } m = n \\ 0 & \text{if } m \neq n \end{cases}$$

Completeness (Closure Relation)

Any arbitrary state vector $|\psi\rangle$ can be expanded as a linear combination of the orthonormal basis kets $|n\rangle$:

$$|\psi\rangle = \sum_n c_n|n\rangle$$

To determine the expansion coefficients c_n, we take the inner product of both sides with a specific basis bra $\langle m|$:

$$\langle m|\psi\rangle = \langle m|\sum_n c_n|n\rangle$$

Using the linearity of the inner product, the summation can be brought outside:

$$\langle m|\psi\rangle = \sum_n c_n \langle m|n\rangle$$

Substituting the orthonormality condition $\langle m|n\rangle = \delta_{mn}$ simplifies the sum:

$$\langle m|\psi\rangle = \sum_n c_n \delta_{mn}$$

Because δ_{mn} is non-zero only when $n = m$, the sum evaluates directly to:

$$c_m = \langle m|\psi\rangle$$

Substituting this expression for the coefficient back into the expansion of $|\psi\rangle$ yields:

$$|\psi\rangle = \sum_n \langle n|\psi\rangle |n\rangle$$

Rearranging the scalar factor $\langle n|\psi\rangle$ to the right of the ket vector $|n\rangle$ gives:

$$|\psi\rangle = \sum_n |n\rangle\langle n|\psi\rangle$$

Factoring out the arbitrary state vector $|\psi\rangle$ from the right-hand side yields:

$$|\psi\rangle = \left(\sum_n |n\rangle\langle n|\right)|\psi\rangle$$

Since this identity must hold for any state vector $|\psi\rangle$, the operator inside the parentheses must be equal to the identity operator $\hat{I}$:

$$\sum_n |n\rangle\langle n| = \hat{I}$$

This fundamental relation is known as the **completeness relation** or **closure relation**.

Expectation Values and Matrix Elements in Bra-Ket Form

Matrix Elements of an Operator

The matrix element of a linear operator $\hat{A}$ between two basis states $|m\rangle$ and $|n\rangle$ is written directly as:

$$A_{mn} = \langle m|\hat{A}|n\rangle$$

Using the completeness relation, an operator $\hat{A}$ can be expressed explicitly in terms of its matrix elements and outer products of the basis states:

$$\hat{A} = \hat{I}\hat{A}\hat{I}$$

Inserting the completeness relation $\hat{I} = \sum_m |m\rangle\langle m|$ on the left and $\hat{I} = \sum_n |n\rangle\langle n|$ on the right gives:

$$\hat{A} = \left(\sum_m |m\rangle\langle m| \right) \hat{A} \left(\sum_n |n\rangle\langle n| \right)$$

Combining the summations together yields:

$$\hat{A} = \sum_m \sum_n |m\rangle\langle m|\hat{A}|n\rangle\langle n|$$

Substituting the definition of the matrix element $A_{mn} = \langle m|\hat{A}|n\rangle$ results in:

$$\hat{A} = \sum_m \sum_n A_{mn}|m\rangle\langle n|$$

Expectation Value of an Operator

In Dirac notation, the expectation value of an observable represented by an operator $\hat{A}$ for a system in a normalized state $|\psi\rangle$ is defined as:

$$\langle A\rangle = \langle\psi|\hat{A}|\psi\rangle$$

We can express this expectation value in terms of the matrix elements A_{mn} by inserting the completeness relation twice. First, writing the expectation value with identity operators inserted:

$$\langle A\rangle = \langle\psi|\hat{I}\hat{A}\hat{I}|\psi\rangle$$

Substituting $\hat{I} = \sum_m |m\rangle\langle m|$ and $\hat{I} = \sum_n |n\rangle\langle n|$ gives:

$$\langle A\rangle = \langle\psi| \left(\sum_m |m\rangle\langle m| \right) \hat{A} \left(\sum_n |n\rangle\langle n| \right) |\psi\rangle$$

Pulling the summations outside the inner product yields:

$$\langle A\rangle = \sum_m \sum_n \langle\psi|m\rangle\langle m|\hat{A}|n\rangle\langle n|\psi\rangle$$

Recognizing that $\langle n|\psi\rangle = c_n$ and $\langle\psi|m\rangle = \langle m|\psi\rangle^* = c_m^*$, we substitute these coefficients into the expression:

$$\langle A\rangle = \sum_m \sum_n c_m^* \langle m|\hat{A}|n\rangle c_n$$

Replacing $\langle m|\hat{A}|n\rangle$ with the matrix element A_{mn} yields the final expanded form for the expectation value:

$$\langle A\rangle = \sum_m \sum_n c_m^* A_{mn} c_n$$

Questions

Short form first (2–3 marks), then long form (5–10 marks). **There are 6 Assam University questions for this unit in the paper bank**, from CHMHCC-502T and CHMDSM-151, marked [AU] with their real source. The other questions are from other sources, marked as such.

Short-form questions

S1. [AU] State Heisenberg's uncertainty principle. — CHMHCC-502T, 5th Sem, 2020, CBCS (Sec B).

Answer in brief: Section 4 — $\Delta x\,\Delta p \geq \hbar/2$; position and momentum cannot both be known sharply at the same time.

S2. [AU] State and explain Heisenberg uncertainty principle. Mention its significance. — CHMDSM-151, 2nd Sem, 2025, FYUG (3 marks).

Answer in brief: Section 4 — $\Delta x\,\Delta p \geq \hbar/2$; it is a property of the wave description, not a measurement defect, and it gives the zero-point energy and the stability of atoms.

S3. [AU] Determine the uncertainty in the momentum of a ball of mass 3x10^-2 kg moving with velocity 600 m/s, if the position can be located with an error of ... — CHMHCC-502T, 5th Sem, 2020, CBCS (Sec A). Note: the position error value is garbled in the paper bank, so use the given value from the question paper.

Answer in brief: Section 4 — $\Delta p \geq \hbar/(2\Delta x)$; insert the given Δx (for example, if $\Delta x = 10^{-3}$ m, $\Delta p \geq 5.3 \times 10^{-32}$ kg m/s).

S4. [non-AU] What is meant by expectation value? — Univ. of Calicut, CBCSS B.Sc 5th Sem, PHY5 B07 Quantum Mechanics, Nov 2021, Q6 (2 marks).

Answer in brief: Section 1 — $\langle A \rangle = \int \psi^* \hat{A} \psi \, dx$, the average of many measurements of A on identically prepared systems.

S5. [standard] What is a stationary state?

Answer in brief: Section 3 — a state of definite energy, $\psi(x,t) = \psi(x)e^{-iEt/\hbar}$; its probability density does not change with time.

S6. [standard] Write the momentum eigenfunction and state its significance.

Answer in brief: Section 6 — in coordinate representation $(1/\sqrt{2\pi\hbar})e^{ip_0 x/\hbar}$; $|\phi(p)|^2 dp$ is the probability of finding momentum between p and $p + dp$.

Long-form questions

L1. [AU] Verify Heisenberg's uncertainty principle for the ground state of the one-dimensional simple harmonic oscillator. — CHMHCC-502T, 5th Sem, 2020, CBCS (Sec B).

What to write: Section 4 in full — for the ground state $\Delta x = \sqrt{\hbar/2m\omega}$ and $\Delta p = \sqrt{m\hbar\omega/2}$, so $\Delta x\,\Delta p = \hbar/2$, the minimum allowed by the principle.

L2. [AU] What is the Heisenberg uncertainty principle? Apply it to the simple harmonic oscillator: zero-point energy and normalized wave function. — CHMHCC-502T, 5th Sem, 2023, CBCS (Sec B).

What to write: Section 4 in full — state $\Delta x\,\Delta p \geq \hbar/2$, minimise $E = (\Delta p)^2/2m + \frac{1}{2}m\omega^2(\Delta x)^2$ to get $E_0 = \frac{1}{2}\hbar\omega$, and write the normalized ground-state wave function $\psi_0(x) = (m\omega/\pi\hbar)^{1/4}e^{-m\omega x^2/2\hbar}$.

L3. [standard] State and prove Ehrenfest theorem, and show how it gives the classical equations of motion for expectation values of position and momentum.

What to write: Section 2 in full — $d\langle x\rangle/dt = \langle p\rangle/m$ and $d\langle p\rangle/dt = -\langle dV/dx\rangle$, proved from the time-dependent Schrodinger equation and its conjugate.

L4. [standard] Explain Dirac bra and ket notation and its operations, and write the Schrodinger wave equation in momentum representation with the momentum eigenfunction and its significance.

What to write: Sections 6–7 in full — kets, bras, inner and outer products, adjoints, orthonormality and completeness; the momentum-representation equation $(p^2/2m + V(i\hbar d/dp))\phi(p) = E\phi(p)$; the momentum eigenfunction and $|\phi(p)|^2$ as a momentum probability density.

Sources

These notes were written from the unit syllabus only, as the user asked. No external PDF, textbook, or lecture notes were used. The questions are from the Assam University paper bank and other universities' papers (see the Questions section above), not from any source used for the notes.